ch13 Wave Motion

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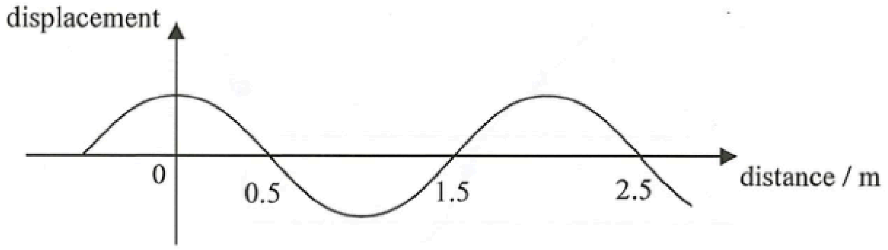
14.

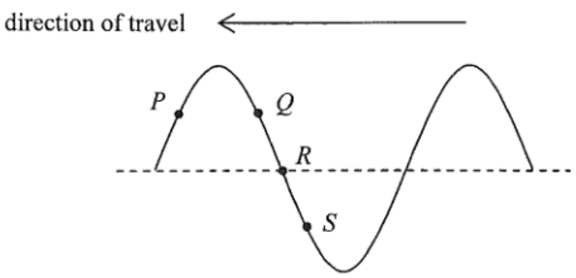


The above figure shows the displacement-distance graph of a wave travelling at a speed of 10 m s^{-1}. What is the period of this wave ?

A. 0.10 s
B. 0.15 s
C. 0.20 s
D. 0.25 s

2014 Q14

14. A transverse wave travels towards the left on a long string. P, Q, R and S are particles on the string. Which of the following statements correctly describe(s) their motions at the instant shown ?

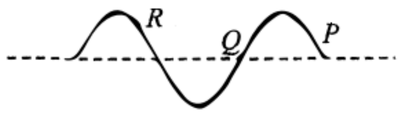


(1) P is moving upwards.
(2) Q and S are moving in opposite directions.
(3) R is momentarily at rest.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2016 Q15

15.



The above figure shows a snapshot of a transverse wave which travels along a string. Which statement is correct ?

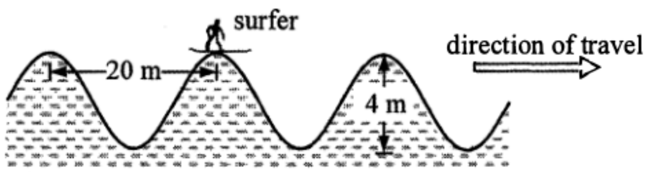
A. The wave is travelling to the left if particle P is moving upwards at this instant.
B. Particles P and R are moving in the same direction at this instant.
C. Particle Q is at rest at this instant.
D. Particle R vibrates with an amplitude larger than that of particle Q.

10. Which of the following can be transferred by mechanical waves from one place to another along the direction of propagation ?

(1) mass
(2) momentum
(3) energy

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

16.

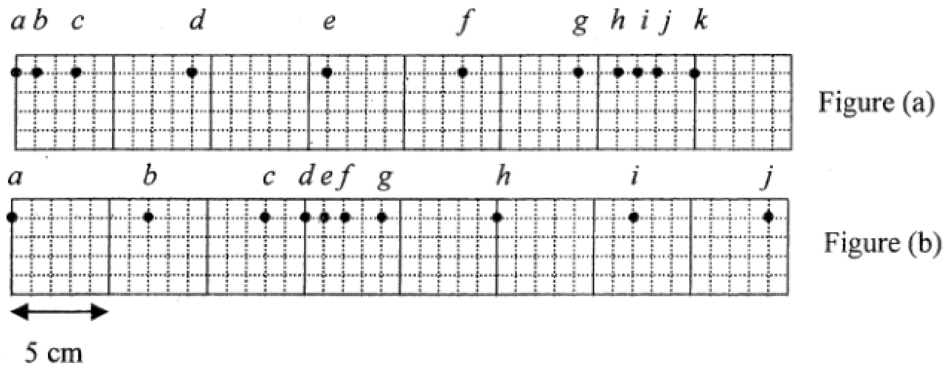


The surfer in the figure reaches a crest at the moment shown. The crests of the water wave are 20 m apart and the surfer descends a vertical distance of 4 m from a crest to a trough in a time interval of 2 s. What is the speed of the wave ?

A. 1 m s^{-1}
B. 2 m s^{-1}
C. 5 m s^{-1}
D. 10 m s^{-1}

2012 Q15

15.



A series of particles is uniformly distributed along a slinky spring initially. Figure (a) shows their positions at a certain instant when a travelling wave propagates along the slinky spring from left to right. Figure (b) shows their positions 0.1 s later. Which statement is correct ?

A. Particle *e* is always stationary.
B. Particles *a* and *i* are in phase.
C. The wavelength of the wave is 16 cm.
D. The frequency of the wave is 10 Hz.

2018 Q14

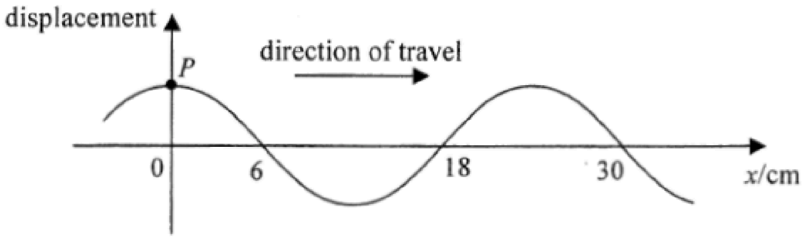
14. Which of the following statements about waves is/are correct ?

(1) Longitudinal waves can transmit energy from one place to another but transverse waves cannot.
(2) Sound waves propagate faster in water than in air.
(3) Infra-red radiation is a kind of electromagnetic wave.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2013 Q16

16.

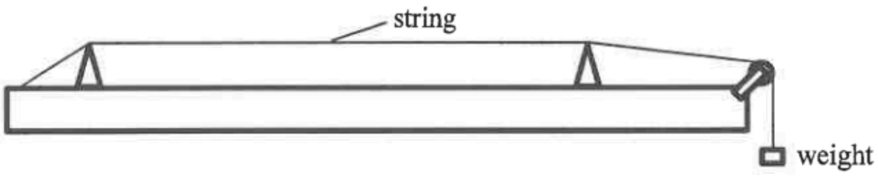


The figure shows a snapshot of a section of a continuous transverse wave travelling along the x-direction at time $t = 0$. At $t = 1.5$ s, the particle P just passes the equilibrium position for a *second* time at that moment. Find the wave speed.

A. 20 cm s^{-1}
B. 12 cm s^{-1}
C. 6 cm s^{-1}
D. 4 cm s^{-1}

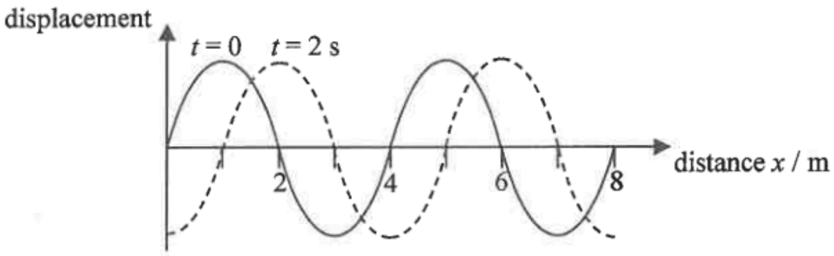
2022 Q16

16. Referring to the set-up below, which of the following combinations would give the highest speed of wave propagation along the string when the string is plucked ?



	tension of the string	radius of the cross-section of the string
A.	T	r
B.	T	$2r$
C.	$2T$	r
D.	$2T$	$2r$

15. The figure shows the displacement-distance graph at time $t = 0$ and $t = 2$ s respectively of a wave travelling to the right.



Which statements about the wave are correct ?

(1) The wavelength is 4 m.
(2) The period is 4 s.
(3) The speed of the wave can be 2.5 m s^{-1}.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2019 Q18

18. How does the speed of propagation of waves along a stretched string change if the tension in the string is increased or the string is replaced by a more massive one of the same length and tension ?

	tension increased	using a more massive string of the same length and tension
A.	speed increases	speed decreases
B.	speed increases	speed increases
C.	speed decreases	speed decreases
D.	speed decreases	speed increases

2021 Q12

12. Earthquakes propagate in the form of waves. The quake centre produces both longitudinal waves (P-wave) and transverse waves (S-wave) which travel with speeds 9.6 km s^{-1} and 6.4 km s^{-1} respectively on the Earth's crust. In an earthquake, a monitoring station detects the arrival of the P-wave pulse at 7:02 a.m. while the S-wave pulse arrives 2 minutes later at 7:04 a.m. Estimate the time that this earthquake occurs.

A. 6:53 a.m.
B. 6:56 a.m.
C. 6:58 a.m.
D. 6:59 a.m.

2013 Q17

17.

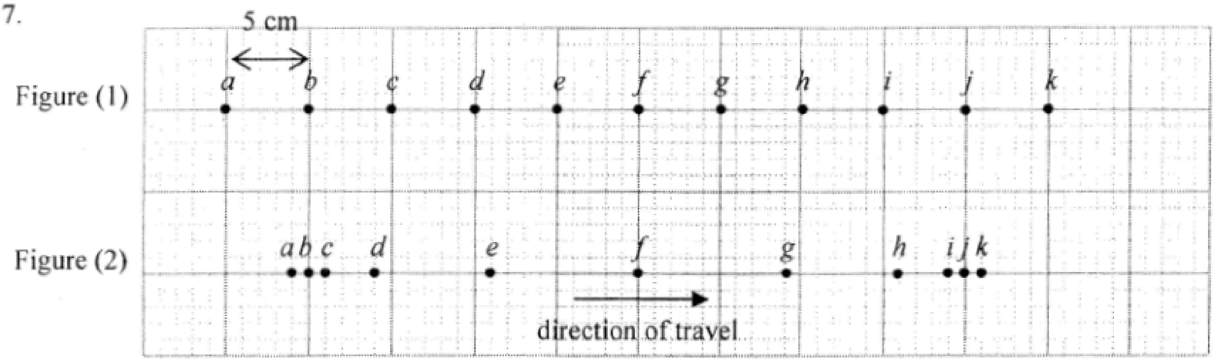
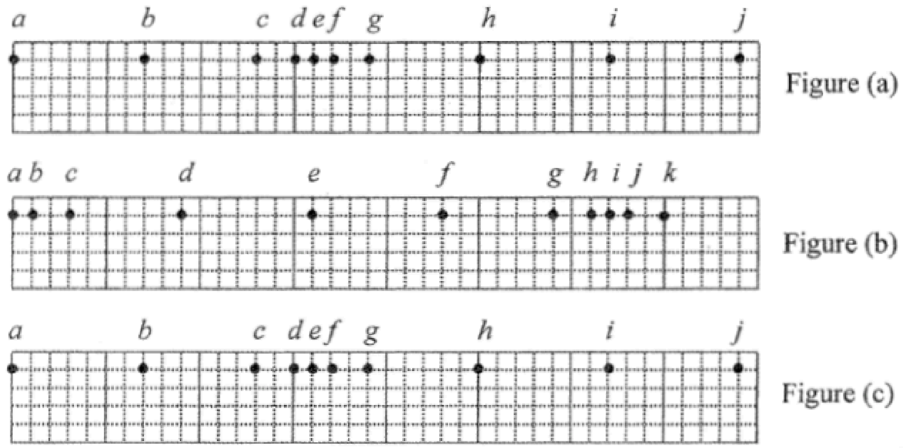


Figure (1) shows the equilibrium positions of particles a to k separated by 5 cm from each other in a medium. A longitudinal wave is travelling from left to right with a speed of 80 cm s^{-1}. At a certain instant, the positions of the particles are shown in Figure (2). Determine the amplitude and frequency of the wave.

	amplitude	**frequency**
A.	6 cm	2 Hz
B.	6 cm	4 Hz
C.	9 cm	2 Hz
D.	9 cm	4 Hz

2015 Q12

12.



A series of particles is uniformly distributed along a slinky spring initially. When a travelling wave propagates along the slinky spring from left to right, Figure (a) shows the positions of the particles at a certain instant. Figures (b) and (c) respectively show their positions 0.05 s and 0.1 s later. Which of the following is/are a possible frequency of the wave ?

(1) 10 Hz
(2) 20 Hz
(3) 40 Hz

A. (1) only
B. (2) only
C. (3) only
D. (1), (2) and (3)

2020 Q11

11.

Figure (1)

Figure (2)

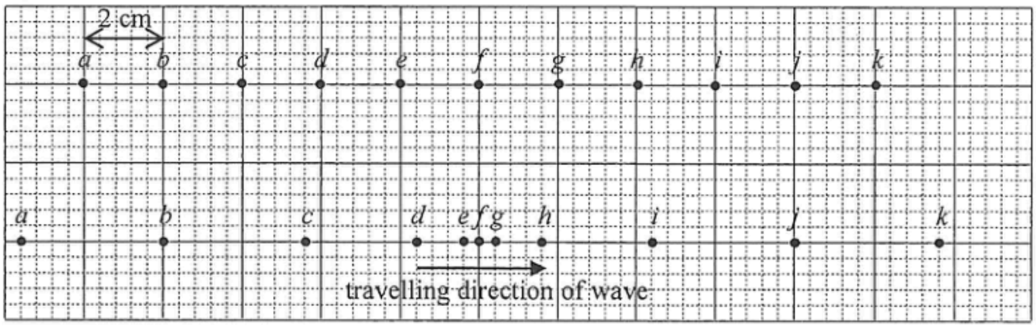
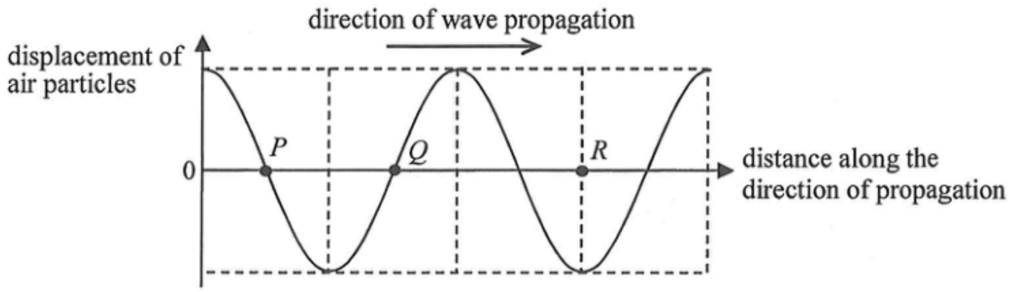


Figure (1) shows the equilibrium positions of particles a to k in a medium. The particles are separated by 2 cm from each other. A longitudinal wave of frequency 5 Hz is travelling from left to right. At a certain instant, the positions of the particles are shown in Figure (2). Determine the amplitude and speed of the wave.

	amplitude	speed
A.	3.6 cm	40 cm s^{-1}
B.	3.6 cm	80 cm s^{-1}
C.	2.4 cm	40 cm s^{-1}
D.	2.4 cm	80 cm s^{-1}

2025 Q15

15. The graph shows the variation of the displacement of air particles from their equilibrium positions with distance for a sound wave at a certain instant. Displacement along the direction of wave propagation is taken to be positive.

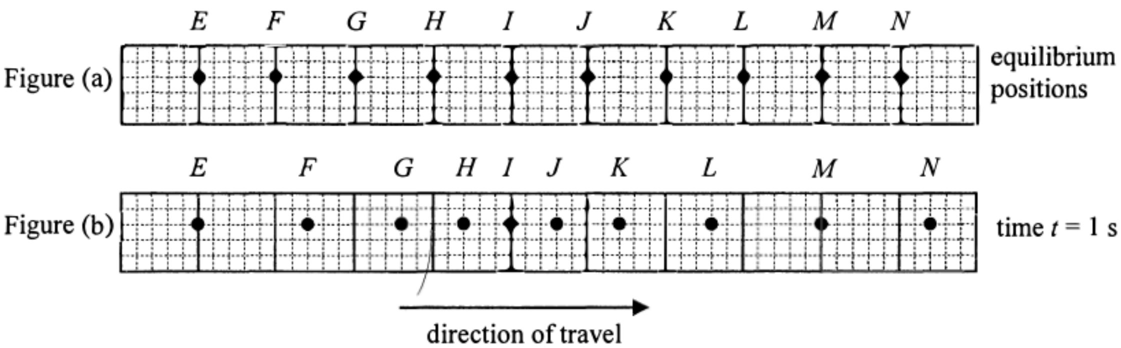


Which statement concerning the wave at that instant is correct ?

A. An air particle at P is momentarily at rest.
B. A rarefaction occurs at Q.
C. A compression occurs at R.
D. An air particle at R is moving with the greatest speed.

2017 Q14

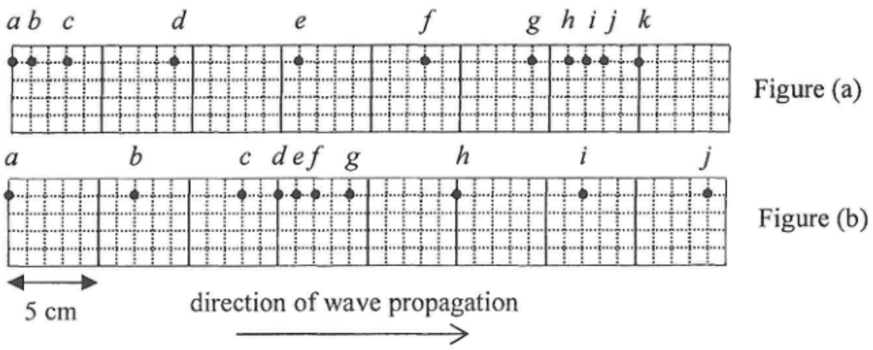
14. Figure (a) shows the equilibrium positions of particles E to N in a medium. At time t = 0, a longitudinal wave starts travelling from left to right. At time t = 1 s, the positions of the particles are shown in Figure (b).



Which of the following statements **MUST BE** correct ?

A. The distance between particles F and N is equal to the wavelength of the wave.
B. The period of the wave is 1 s.
C. Particle E is always at rest.
D. Particle I is momentarily at rest at t = 1 s.

16.

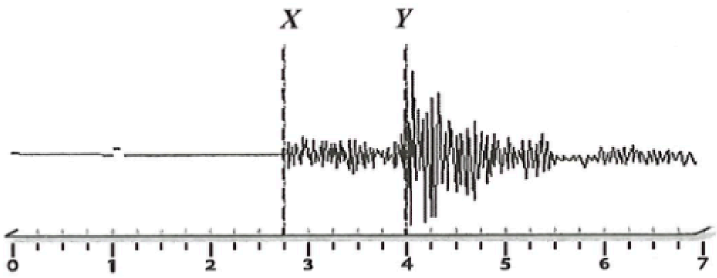


A series of particles is uniformly distributed along a slinky spring initially. Figure (a) and (b) show their positions at two instants when a travelling wave of frequency 10 Hz propagates along the spring from left to right. Which statement below is correct ?

A. Particle *a* is always stationary.
B. Particle *b* and *c* vibrate with different amplitudes.
C. The wavelength of the wave is 16 cm.
D. The smallest time interval for obtaining these two figures is 0.05 s.

2023 Q15

15. Earthquakes produce P-waves (longitudinal) and S-waves (transverse) which travel at 7.0 km s^{-1} and 4.0 km s^{-1} respectively. On a given day, a seismometer detected the P- and S- waves from a shallow earthquake (see the figure below) with its quake centre located at a distance *D* from the seismometer.

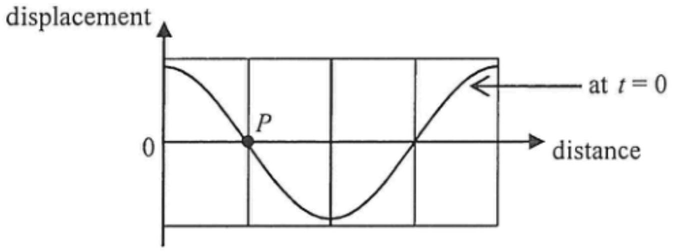


Which deduction below is correct ?

	Y in the figure denotes the arrival of	*D* / km
A.	P-waves	225
B.	P-waves	700
C.	S-waves	225
D.	S-waves	700

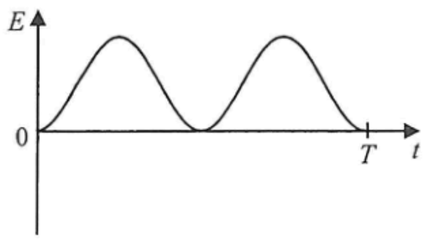
2020 Q12

12. The figure shows part of the displacement-distance graph of a travelling wave of period T at time $t = 0$. P is a particle on the wave.

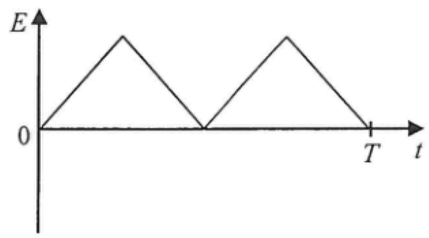


Which graph below correctly shows the variation of the particle's kinetic energy E within a period starting from $t = 0$?

A.

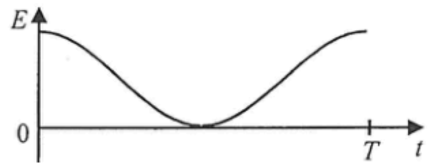


B.



C.

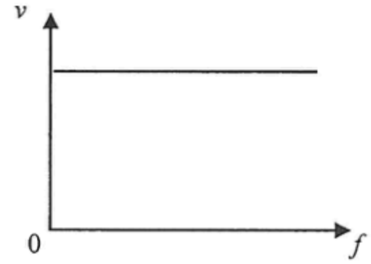
D.



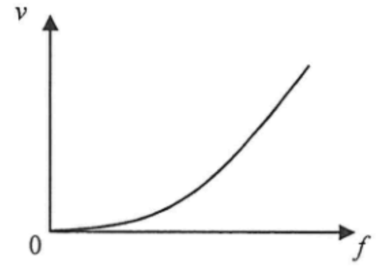
2020 Q16

16. A transverse wave propagates along a stretched string. Which graph below correctly shows the variation of the speed v of the wave with its frequency f?

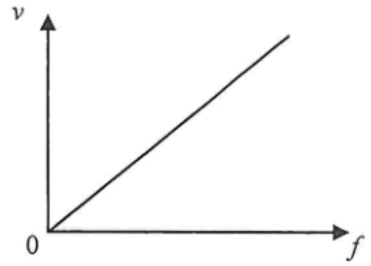
A.



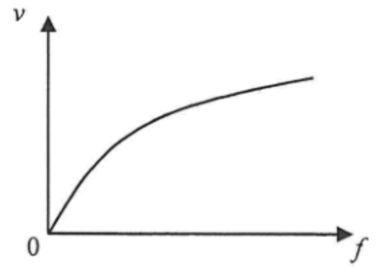
B.



C.

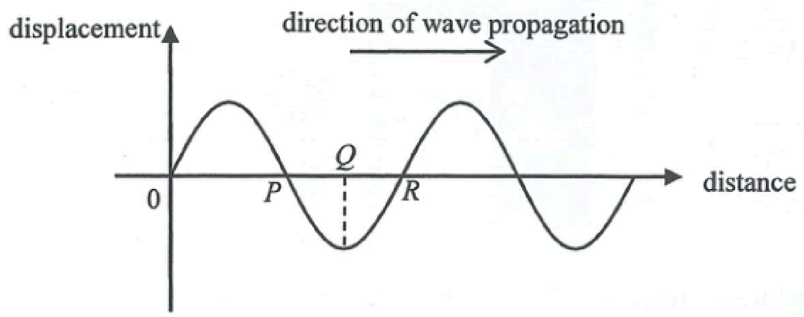


D.



2019 Q14

14. The figure shows the displacement-distance graph at a certain instant of a longitudinal wave which travels to the right. Displacement to the right is taken to be positive.



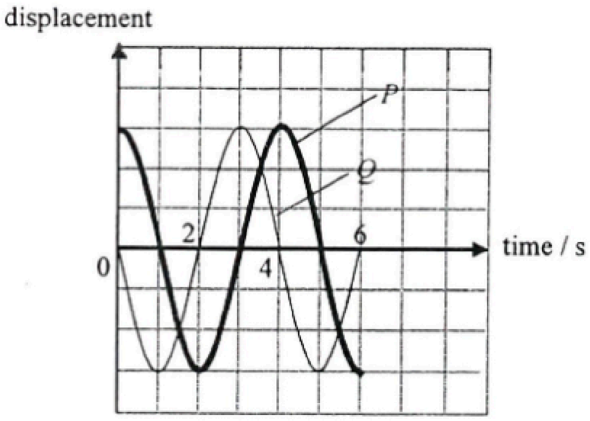
At the instant shown, which of the following statements is/are correct ?

(1) P is a centre of compression.
(2) A particle with its equilibrium position at Q is at rest.
(3) A particle with its equilibrium position at R is moving downwards.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2018 Q15

15. The figure below shows the displacement-time graph of particles P and Q on the same transverse travelling wave of wavelength λ.



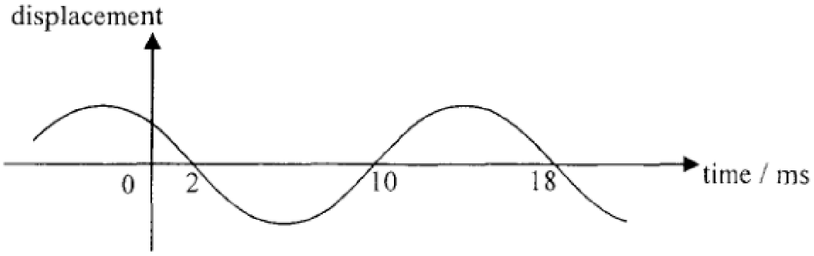
Which of the following statements **MUST BE** correct ? Upward displacement is taken to be positive.

(1) At time $t = 2$ s, P is momentarily at rest.
(2) At time $t = 4$ s, Q is moving downwards.
(3) The separation between the equilibrium positions of P and Q is 0.25λ.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

2021 Q11

11.



The displacement-time graph of a particle on a travelling wave is as shown. Find the frequency of the wave.

A. 55.6 Hz
B. 62.5 Hz
C. 111 Hz
D. 125 Hz

2026 Q15

15.

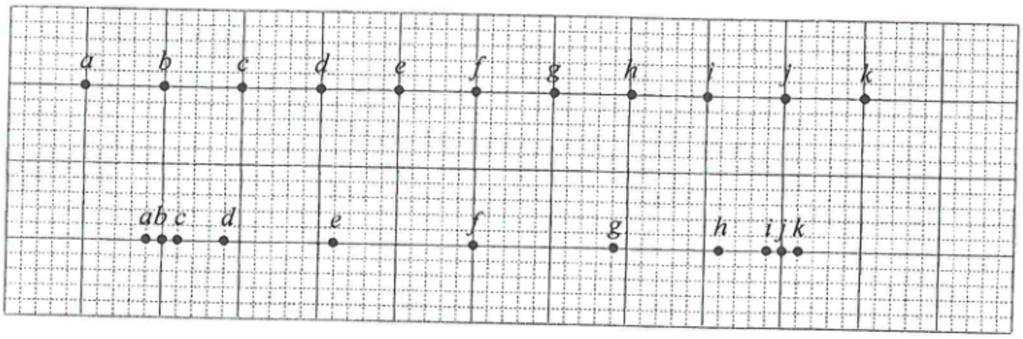


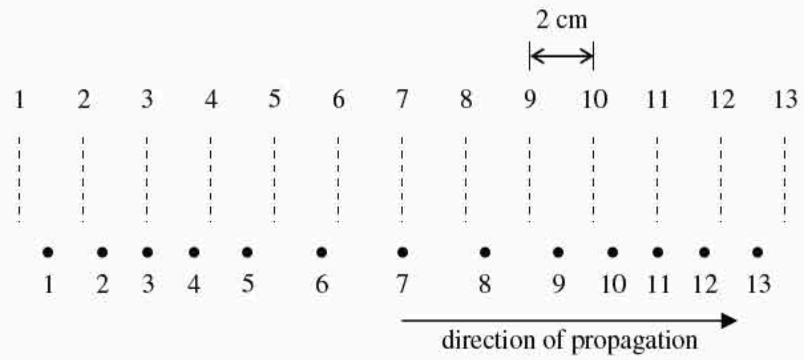
Figure (1) shows the equilibrium positions of particles a to k in a medium. A longitudinal wave is travelling through the medium and the positions of the particles at a certain instant are shown in Figure (2). Which of the following statements is/are correct at that instant ?

(1) The separation between c and k is one wavelength.
(2) f is momentarily at rest.
(3) i and k are moving in the same direction.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

pp Q17

17.



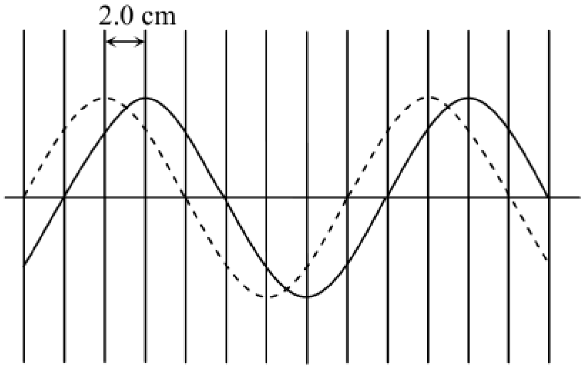
A longitudinal wave travels to the right through a medium containing a series of particles. The figure above shows the positions of the particles at a certain instant. The dotted lines indicate the equilibrium positions of the particles. Which of the following statements about the wave at the instant shown is/are correct ?

(1) The wavelength of the longitudinal wave is 16 cm.
(2) Particles 8 and 10 are moving in the same direction.
(3) Particle 3 is momentarily at rest.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

sap Q17

17.

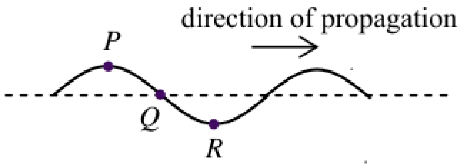


The solid curve in the diagram shows a transverse wave at a certain instant. After 0.05 s, the wave has travelled a distance of 2.0 cm and is indicated by the dashed curve. Find the wavelength and frequency of the wave.

	Wavelength/cm	**Frequency/Hz**
A.	8	2.5
B.	16	2.5
C.	8	5
D.	16	5

sap Q18

18.



The figure shows the shape of a transverse wave travelling along a string at a certain instant. Which statement about the motion of the particles *P*, *Q* and *R* on the string at this instant is correct ?

A. Particle *P* is moving downwards.
B. Particle *Q* is stationary.
C. Particle *R* attains its maximum acceleration.
D. *P* and *Q* are in phase.